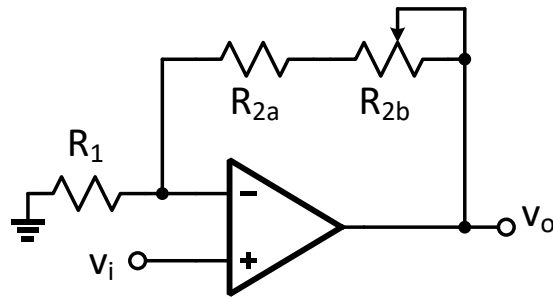


LAB 5 – LINEAR OSCILLATORS

Introduction

In this lab, you will build and test several op-amp based linear oscillators.

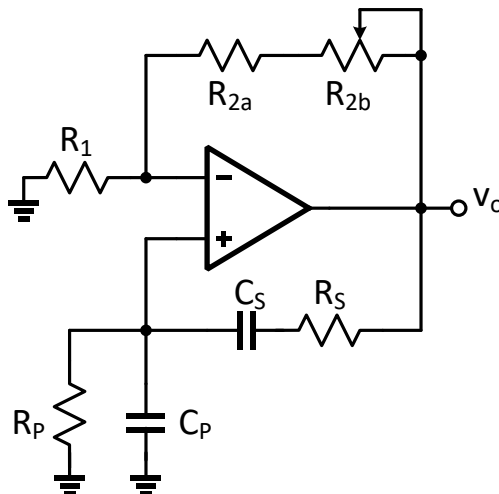
Experiment 1. Wein-Bridge Oscillator



The amplifier shown above uses the LMC660 op amp with ± 8 V supplies. All other component values are given in the table below.

Component	Value
R_1	$15\text{ k}\Omega$
R_{2a}	$22\text{ k}\Omega$
R_{2b}	$10\text{ k}\Omega$ Potentiometer

1. Build the circuit. The feedback resistance, R_2 , is formed by the series combination of the $22\text{ k}\Omega$ fixed resistor and the $10\text{ k}\Omega$ potentiometer.
2. Test the operation of the amplifier using a sinusoidal input with a frequency of 1,000 Hz. Adjust the potentiometer so that the gain is in the range of $2.5 - 2.75\text{ V/V}$.



We will now turn the simple amplifier into an oscillator by adding the Wein-bridge tank circuit shown above. Component values for the tank are given in the table below.

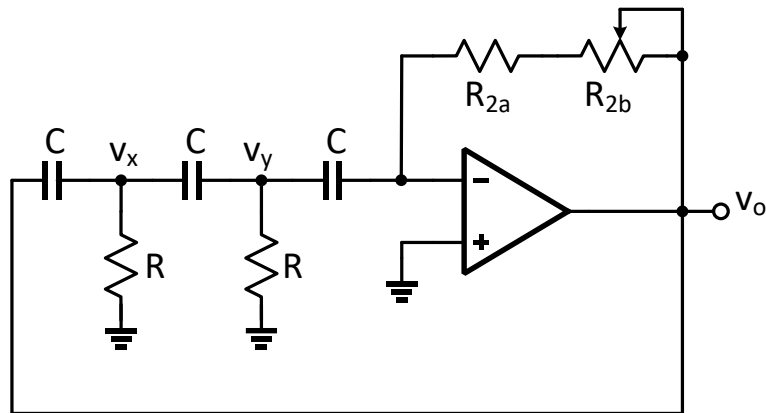
Component	Value
R_P	$10\text{ k}\Omega$
C_P	10 nF
R_S	$10\text{ k}\Omega$
C_S	10 nF

3. Observe the output voltage on the oscilloscope. Initially, the output will probably be a flat line at 0 V. Slowly adjust the potentiometer to increase the gain of the amplifier. When the amplifier gain increases to a value of 3 V/V, the circuit should become unstable and begin oscillating. Try to adjust the potentiometer so that the output is not clipped. Save a copy of the oscilloscope trace for your lab report.
4. Measure the oscillation frequency. Confirm that the frequency matches the expected value of $\omega_{osc} = 1/R_P C_P = 1/R_S C_S$.
5. Adjust the potentiometer to its maximum value. The output oscillation should be clipped. Save a screen shot of the distorted waveform for your lab report.
6. Measure the oscillation frequency of the clipped waveform and compare it to the value measured previously.

Experiment 2. Wien-Bridge Oscillator 2

1. Design the Wein-bridge oscillator from Experiment 1 to have an oscillation frequency of 5 kHz ($\pm 10\%$).
2. Include the schematic of your design along with all derivations in your lab report.
3. Use LTSpice to perform a transient simulation of your proposed design showing the expected frequency of oscillation. You may need to experiment with the value of the feedback resistor to get the simulation to not clip. Include a screen shot of the simulation output in your lab report.
4. Build the circuit and adjust the gain so that the output is sinusoidal without clipping.
5. Measure the oscillation frequency.
6. Include a screen shot of the output for your report.

Experiment 3. Phase-Shift Oscillator

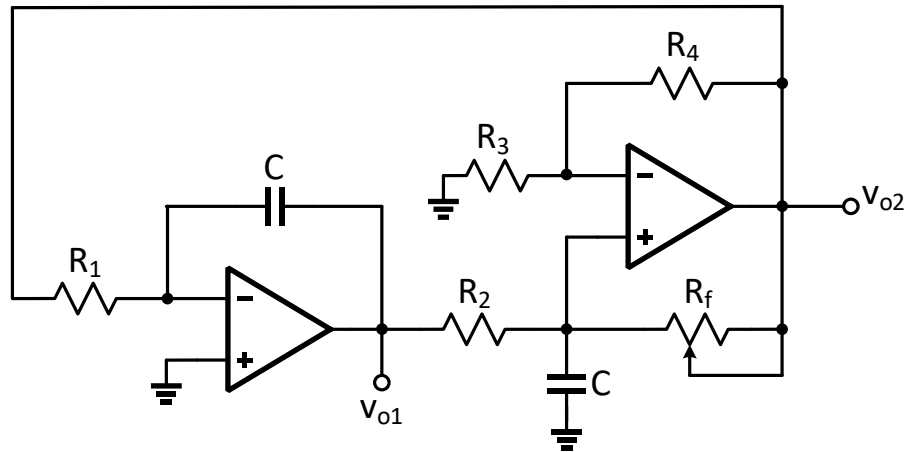


The above oscillator uses the LMC660 op amp with ± 8 V supplies. All other component values are given in the table below.

Component	Value
R	$10\text{ k}\Omega$
C	10 nF
R_{2a}	$100\text{ k}\Omega$
R_{2b}	$100\text{ k}\Omega$ Potentiometer

1. Build the circuit and adjust the potentiometer until the circuit begins oscillating and the output is undistorted. Save a copy of the oscilloscope trace for your lab report.
2. Measure the oscillation frequency and confirm that it matches the expected value.
3. Adjust the potentiometer to its maximum value. The output signal should be clipped. Save a screen shot of the distorted waveform for your lab report.
4. Measure the oscillation frequency of the clipped waveform and compare it to the value measured previously.

Experiment 4. Quadrature Oscillator



The above oscillator uses the LMC660 op amp with ± 8 V supplies. All other component values are given in the table below. The value of R_1 can be achieved using two $2.2\text{ k}\Omega$ resistors in parallel or using a $1\text{ k}\Omega$ resistor and a $100\text{ }\Omega$ resistor connected in series.

Component	Value
R_1	$1.1\text{ k}\Omega$
R_2	$2.2\text{ k}\Omega$
R_3	$2.2\text{ k}\Omega$
R_4	$2.2\text{ k}\Omega$
R_f	$10\text{ k}\Omega$ Potentiometer
C	100 nF

1. Build the circuit. The potentiometer, R_f , must be set to approximately $2.2\text{ k}\Omega$, or less, to have sufficient gain for the circuit to oscillate. Adjust the potentiometer until the circuit begins to oscillate without any clipping in the outputs.
2. Display both v_{o1} and v_{o2} together on the oscilloscope. They should be 90° out of phase. Save a screen shot of the waveform for your lab report.
3. Further decrease the value of the potentiometer, R_f , to distort the waveforms. Note that v_{o2} will distort sooner and be worse than v_{o1} . Save a screen shot of the distorted waveform for your lab report.